Kipi System: Agent Pipeline Architecture

25 agents across 9 phases. Bus/JSON communication. No shared context. Every step logged, every failure handled.

V1.0 ARCHITECTURE

PIPELINE ORCHESTRATION

PHASE 0

Preflight — System Gate

Single Agent · Sequential

Before any data flows, one agent verifies the entire toolchain. If any critical connection is unavailable, the pipeline halts immediately — no partial runs, no silent failures.

MCP Connections Checked

- Calendar
- Gmail
- CRM
- Browser

Gate condition: `preflight.json` must return `ready=true` before Phase 1 begins.

PHASE 1

Data Ingest — 4 Agents in Parallel

All four agents fire simultaneously, each pulling from an independent source and writing structured JSON output to the `bus/` directory.

Calendar Pull

Fetches today's meetings, attendees, and scheduling context.

Gmail Pull

Reads priority inbox threads and extracts action signals.

CRM Pull

Loads contact records, deal stages, and pipeline history.

External Pipeline Pull

Ingests external lead and prospect data into the bus.

PHASE 2

Analysis — 2 Independent Agents

🧠 Meeting Prep Agent

Cross-references calendar data with CRM records to generate contextual briefings for every meeting scheduled today. Outputs structured prep notes to bus.

- Reads: `calendar.json` + `crm.json`
- Writes: `meeting_prep.json`

🔗 Warm Intro Match Agent

Scans the external pipeline against existing CRM contacts, surfacing warm introduction opportunities and mutual connections.

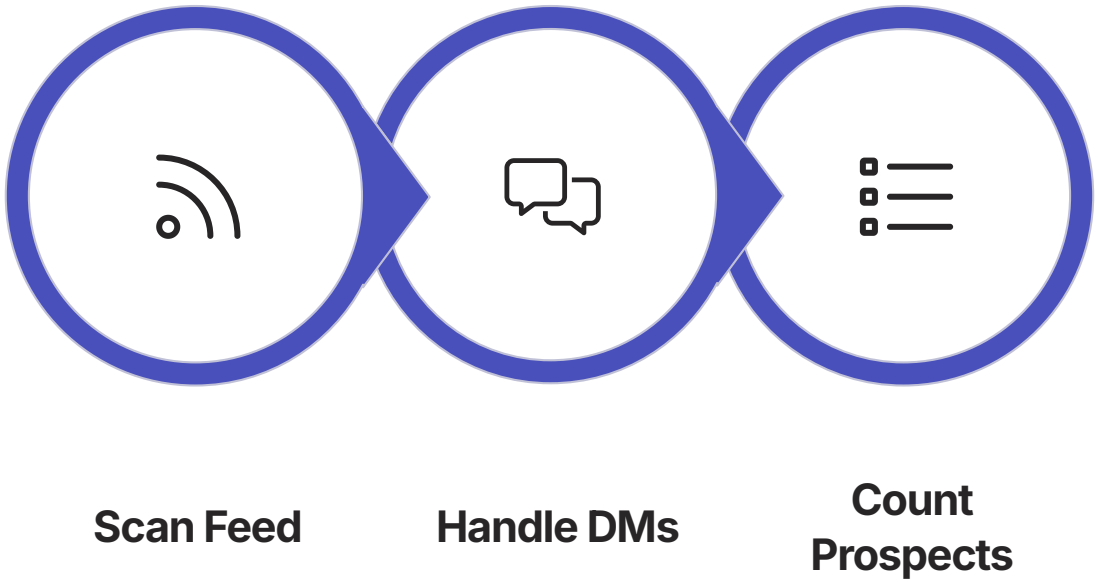
- Reads: `pipeline.json` + `crm.json`
- Writes: `warm_intros.json`

Both agents run **simultaneously** — fully independent, no shared state or cross-dependency.

PHASE 3

Social — Sequential Browser Agents

The browser can only hold **one active tab at a time**, so these three agents must run in strict sequence. Each hands off control before the next begins.



Output feeds directly into Phase 5's pipeline scoring, providing fresh social context for lead temperature calculations.

PHASE 4

Content — Sequential Then Parallel

Execution Order

Signals agent must complete first — it feeds the parallel agents below. On brand days, the brand post agent runs before all others.

Brand day rule: Brand Post agent gets priority execution before the standard content queue.

01

Signals Agent

Scans news sources, identifies trending topics, drafts initial post copy.

02

Value Routing *(parallel)*

Routes content to appropriate audience segments and channels.

03

Post Visuals *(parallel)*

Generates visual assets to accompany finalized post copy.

PHASE 5

Pipeline — 4 Parallel + Conditional Fallback

1	2	3	4
Temperature Scoring	**Lead Sourcing**	**Follow-Up Agent**	**Loop Review**
Ranks every lead by engagement recency and deal signals.	Finds new prospects via primary sources. **On failure → auto-fallback to browser scraping.**	Drafts follow-up messages for dormant and active threads.	Audits previous cycle outputs for unresolved action items.

Engagement Hitlist: After all 4 complete, a final agent reads all bus files and produces copy-paste ready action output — the daily action surface.

PHASE 6–7

Compliance + Synthesis

PHASE 6 · COMPLIANCE

2 Agents in Parallel

Both compliance agents independently review all content outputs against brand positioning rules and communication guidelines before anything is published or acted upon.

- Content compliance check
- Positioning rules validation
- Flags violations before write-back

PHASE 7 · SYNTHESIS

1 Agent · Full Bus Reader

The synthesis agent reads **every single bus file** produced across all prior phases and consolidates them into one authoritative output.

- Reads all `bus/*.json` files
- Produces `daily_schedule.json`
- Single source of truth for Phase 8

PHASE 8–9

Build + Write-Back

PHASE 8 · BUILD

Build Script

Converts `daily_schedule.json` into a rendered HTML daily brief.

Visual Verify

Opens rendered output in browser for human-readable confirmation.

Audit Harness

Checks for any skipped phases, missing outputs, or incomplete steps.

PHASE 9 · WRITE-BACK

2 Agents · Externalize

The final two agents push the day's work back into the world — no data stays siloed in the bus.

- **CRM Push Agent:** writes actions, notes, and stage updates back to CRM
- **Checklist Agent:** updates daily operational checklists with completion status

Key Design Principles

The architecture is opinionated by design. These five rules govern every agent's behavior across every phase.

File-Based Communication

Agents share zero context. All communication happens through `bus/` JSON files — clean, auditable, decoupled.

Daily Overwrite

Bus files reset every run. No stale state bleeds between sessions. Each day starts from zero.

Automatic Fallbacks

Failure chains trigger predefined fallback routes. Lead sourcing failure → browser scraping. No manual intervention required.

Every Step Logged

Full execution logs are written at each phase. The audit harness validates completeness before write-back proceeds.

No Silent Skips

The AI cannot skip a step without surfacing an explicit prompt. Gates enforce execution order — the system is deterministic by design.